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To cite this article: Manoj Dhondiram Patil, K. Vadirajacharya & Swapnil W. Khubalkar (2021): Design and Tuning of Digital Fractional-Order PID Controller for Permanent Magnet DC Motor, IETE Journal of Research, DOI: [10.1080/03772063.2021.1942243](https://doi.org/10.1080/03772063.2021.1942243)

To link to this article: <https://doi.org/10.1080/03772063.2021.1942243>



Published online: 28 Jun 2021.



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Design and Tuning of Digital Fractional-Order PID Controller for Permanent Magnet DC Motor

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ABSTRACT

One of the central problems in control theory is related to the design of controllers for the improvement of system performance. The aim of this paper is to design an efficient digital fractional-order proportional-integral-derivative (FO-PID) controller for the speed control of buck converter fed permanent magnet DC (PMDC) motor. Speed control is achieved by a pulse width modulated control. The FO-PID controller parameters (gain and order) are obtained by using the ant colony optimization (ACO) technique. The effectiveness of the proposed control scheme is simulated and verified using MATLAB/Simulink results. The performance comparison shows that the speed control of buck converter fed PMDC motor is improved with proposed FO-PID controller than that of integer-order PID controller.

KEYWORDS

Buck converter;
Fractional-order PID;
Optimization; PMDC motor
expression; Speed control

1. INTRODUCTION

The industrial sector has increased the demand for process automation in many areas [1]. Automation results in the increased production, improved quality, and reduced cost. The variable speed drives, until a couple of decades back, had various limitations such as poor efficiencies and large sizes. However, the advent of power electronic circuits led to better conditions, and now variable drive systems are available which are not only smaller in size but also have higher efficiency and reliability [2].

The PMDC motor is broadly used in control systems for analytical purposes. It is necessary to establish mathematical models for DC motors in different applications. Speed control of the PMDC motor is required for a large number of industrial and domestic applications such as those used in the robotic applications, electric vehicles, transportation systems, textile mills, servo systems, etc. The speed of PMDC motor can be controlled by armature voltage control, proportional-integral (PI) controller, proportional-integral-derivative (PID) controller, and intelligent controllers. These controllers use artificial intelligence (AI) computing techniques like evolutionary algorithms, neural networks, and fuzzy logic [2].

A DC to DC buck converter allows the smooth control of a PMDC motor by applying the required voltage. The buck converter has two energy storing elements, *i.e.* an inductor and a capacitor which generates smooth output

voltages and currents [3]. Fadil *et al.* proposed a controller based on the backstepping technique for speed control of a buck converter fed DC motor [4]. Ahmad *et al.* presented a comparison of the PI and LQR (linear quadratic regulator) controllers for a buck converter fed DC motor [5]. An H-infinity controller, with pole clustering based on linear matrix inequalities techniques for speed control of a DC motor fed by a buck converter, was presented in Mohd Tumari *et al.* [6]. Simplified predictive control (SPC) and shifted dynamic matrix control (SDMC) have been attempted on a PMDC motor drive [7]. Literature related to the control of power converter fed DC motor has been reported in [3,8].

The PID-based control technique is preferred in many applications because of its simple structure. It has advantages like faster control and reduced settling time [9]. Recently, PID control schemes have been extended to their generalized form using fractional calculus, involving integration and differentiation of a fractional order [10–15]. Thus, in order to achieve the desired speed response, this paper focuses on the design and tuning of an FO-PID controller for buck converter fed PMDC motor.

The applications of fractional calculus have been reported in the literature [16,17] and its advantages are proven as well [18–20]. The FO-PID controller has fractional differ-integrator operations. This controller has a memory (*i.e.*

its action memorizes its past states) and hence it avoids instantaneous actions. Fractional order controllers give more flexibility in adjusting the gain and phase characteristics than integer-order controllers [11].

The main contribution of this paper is to propose a digital FO-PID controller that carries out the smooth speed control task for buck converter fed PMDC motor. Fine tuning of controller parameters is required to design an optimum controller, which in turn calls for real parameter optimization in five-dimensional hyperspace [21–24]. Here, the ACO optimization technique is used for tuning the parameters of FO-PID controller. The validation of the proposed controller performance for the speed control of PMDC motor is demonstrated. As far as research concern, there is very less literature available on the applications of FO-PID controller. This paper provides a detailed idea of design and tuning.

This work is organized as follows. Section 2 gives the modeling of the system. The design, approximation, and tuning details of the proposed digital FO-PID controller are given in Section 3. In Section 4, the detailed simulation results are provided. The paper concluded in Section 5.

2. MATHEMATICAL MODELING OF PMDC MOTOR DRIVE

Figure 1 presents the block diagram of closed-loop speed control of buck converter fed PMDC motor. In this control scheme, FO-PID controller is considered for the speed control of buck converter fed PMDC motor drive. A reference speed is given as set point for the controller, which provides the control signal computing the speed error generated in the control scheme. The PWM pulses are generated at a particular duty cycle corresponding to the controller output. This signal operates the switch of the buck converter and regulates the armature voltage of PMDC motor accordingly speed control is achieved. The voltages applied to the motor are represented by the input supply. The resistance and inductance of the armature of

the motor are represented by R_a and L_a . The armature is modeled as a circuit with a resistance connected in series with an inductance. The PMDC motor is controlled by changing the armature voltage.

$$\frac{\omega(s)}{V_a(s)} = \frac{K_t}{J_m L_a s^2 + (J_m R_a + B_m R_a + B_m L_a)s + K_t^2} \quad (1)$$

Equation (1) is the overall transfer function for speed control of the PMDC motor which infers that with a change in armature voltage speed of the motor can be controlled. Parameters of the PMDC motor are as follows: resistance $R_a = 3.84 \text{ } (\Omega)$, inductance $L_a = 0.035 \text{ (H)}$, torque constant $K_t = 1.8 \text{ (N} \cdot \text{m/A)}$, friction coefficient $B_m = 0.015 \text{ (N} \cdot \text{m} \cdot \text{s/rad)}$, and moment of inertia $J_m = 0.045 \text{ (kg} \cdot \text{m}^2)$. The buck converter is a power electronic circuit and is known as switching regulators. They produce an output voltage between 0 (zero) and input voltage. The duty cycle variation of pulse width modulated (PWM) signal regulates the voltage across armature winding of the PMDC motor [3–8]. Figure 2 shows the design of the buck converter circuit. The state-space representation of buck converter is given in (2)–(3) [27].

$$\begin{bmatrix} \frac{di_L}{dt} \\ [6pt] \frac{dv_o}{dt} \end{bmatrix} = \begin{bmatrix} -\frac{R_L}{L} & -\frac{1}{L} \\ [6pt] \frac{1}{C} & -\frac{1}{RC} \end{bmatrix} \begin{bmatrix} I_L \\ v_o \end{bmatrix} + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} v_{in} \delta \quad (2)$$

where I_L is a load current and V_0 is a output voltage. The corresponding transfer function is given as,

$$G(s) = \frac{v_o(s)}{\delta(s)} = \frac{v_{in}}{as^2 + bs + c} \quad (3)$$

where $a = LC$, $b = \frac{L}{R} + R_L C$, $c = 1 + \frac{R_L}{R}$, load resistance $R = 3.84 \text{ } (\Omega)$, inductance $L = 0.04215 \text{ (H)}$, capacitance $C = 1000 \text{ } (\mu\text{F})$, inductance resistance $R_L = 1.1 \text{ } (\Omega)$, input voltage $v_{in} = 24 \text{ (V)}$.

A permanent magnet DC motor mainly consists of permanent magnets for the field and armature winding has

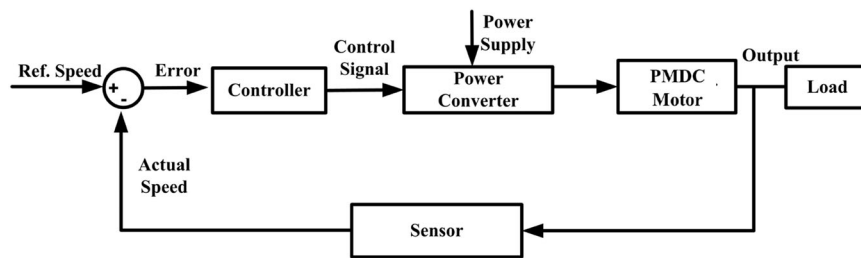


Figure 1: Block diagram of speed control system

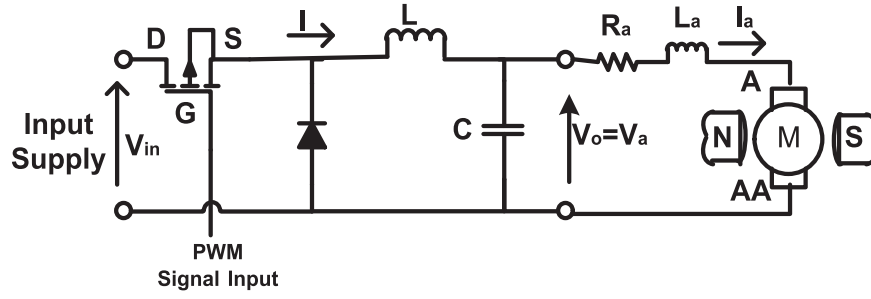


Figure 2: Schematic diagram of buck converter fed PMDC motor

current-carrying conductors [25]. Figure 2 shows the schematic diagram of the buck converter fed PMDC motor. The general transfer function of the PMDC motor is shown in (1) [26].

3. PROPOSED DESIGN OF DIGITAL FO-PID CONTROLLER

Basic definitions of fractional calculus and approximation of fractional differentiator and integrator are reported in the literature [10,11]. The integer-order operator can be generalized as follows,

$${}_a D_t^\alpha = \begin{cases} \frac{d^\alpha}{dt^\alpha} & \alpha > 0 \\ 1 & \alpha = 0 \\ \int_a^t (d\tau)^{-\alpha} & \alpha < 0 \end{cases} \quad (4)$$

The Riemann–Liouville definition for the general fractional integrals or derivatives is given in (5) as,

$${}_a D_t^\alpha f(t) = \frac{1}{\Gamma(n-\alpha)} \left(\frac{d}{dt} \right)^n \int_a^t \frac{f(\tau)}{(t-\tau)^{\alpha-n+1}} d\tau \quad (5)$$

where α is a real number, n is an integer, and $\Gamma(\cdot)$ is the Euler's gamma function. The Laplace transform of (5) is presented in (6) as,

$$L\{{}_a D_t^{\pm\alpha} f(t)\} = s^{\pm\alpha} F(s) \quad (6)$$

A fractional-order system is described by the fractional differential equation (7) as,

$${}_a D^{\alpha n} f(t) + {}_a D^{\alpha n-1} f(t) + \dots = D^{\beta n} f(t) + D^{\beta n-1} f(t) + \dots \quad (7)$$

where $D^{\alpha n}$ is a fractional derivative of order αn with respect to variable t . The generalized PID controller is

given by the differential equation (8) as,

$$u(t) = K_p e(t) + K_i D^{-\alpha} e(t) + K_d D^{\beta} e(t) \quad (8)$$

Using Laplace transform, the transfer function of the controller (8) can be expressed as (9),

$$u(s) = K_p + K_i s^{-\alpha} + K_d s^{\beta} \quad (9)$$

where K_p , K_i , and K_d are the proportional, integral, and derivative constant, α, β are positive real numbers, $e(t)$ gives the deviation between the actual value and specified value.

3.1 Approximation Method

The fractional-order operators are of infinite-dimensional nature. It creates hardware implementation issues for the application of FO-PID controllers in the industry [28]. However, a band-limited higher integer-order approximation of the fractional operators gives satisfactory performance [29–32]. Each transfer function can be approximated by its poles and zeros [33–38]. Various methods are available for approximation. Out of which, a method by Oustaloup *et al.* is widely used [34,35]. According to Oustaloup *et al.*, a real non-integer fractional-order operator $(s/\omega_u)^\alpha$ (ω_u is unit gain frequency) can be represented as,

$$D(s) = G(s) = \left(\frac{\omega_L}{\omega_u} \right)^\alpha \left(\frac{1 + s/\omega_L}{1 + s/\omega_H} \right) \quad (10)$$

where ω_L and ω_H are low and high transitional frequencies respectively and $\omega_u = \sqrt{\omega_L \omega_H}$. The high and low transitional frequencies are chosen such that $\omega_L < \omega_A$, and $\omega_H > \omega_B$, where the desired frequency range has been chosen as $\omega_L = 0.1\omega_A$ and $\omega_H = 10\omega_B$. Equation (8) can be expressed as, $D(s) = \lim_{N \rightarrow \infty} D_N(s)$, and

$$D_N(s) = \left(\frac{\omega_u}{\omega_H} \right)^\alpha \prod_{k=-N}^N \left(\frac{1 + s/\omega'_k}{1 + s/\omega_k} \right) \quad (11)$$

where $\omega_k = \omega_L \left(\frac{\omega_H}{\omega_L} \right)^{((k+N+1/2+\alpha/2)/(2N+1))}$ and $\omega'_k = \omega_L \left(\frac{\omega_H}{\omega_L} \right)^{((k+N+1/2-\alpha/2)/(2N+1))}$.

The main point in digital implementation of fractional-order controller is the discretization of fractional-order operator s^α [25,29–31]. For discretization, two steps are required. First one is frequency domain fitting in continuous time and then discretizing fit continuous

transfer function. Here, “Tustin” approximation method with 0.001 (s) as sample time is used for discretization. The Tustin method uses the approximation as given in Equation (11) to relate s -domain and z -domain transfer functions.

$$z = e^{sT_s} \approx \frac{1 + sT_s/2}{1 - sT_s/2} \quad (12)$$

In continuous to discrete time conversions, the discretization $u_d(z)$ of a continuous transfer function $u(s)$

Table 1: Parameters of controllers obtained using ACO

Sr. no.	Controllers	K_p	K_i	K_d	α	β
1	PID	12.6	0.52	0.069	–	–
2	FO-PID	12.06	16.37	0.65	0.6	0.5

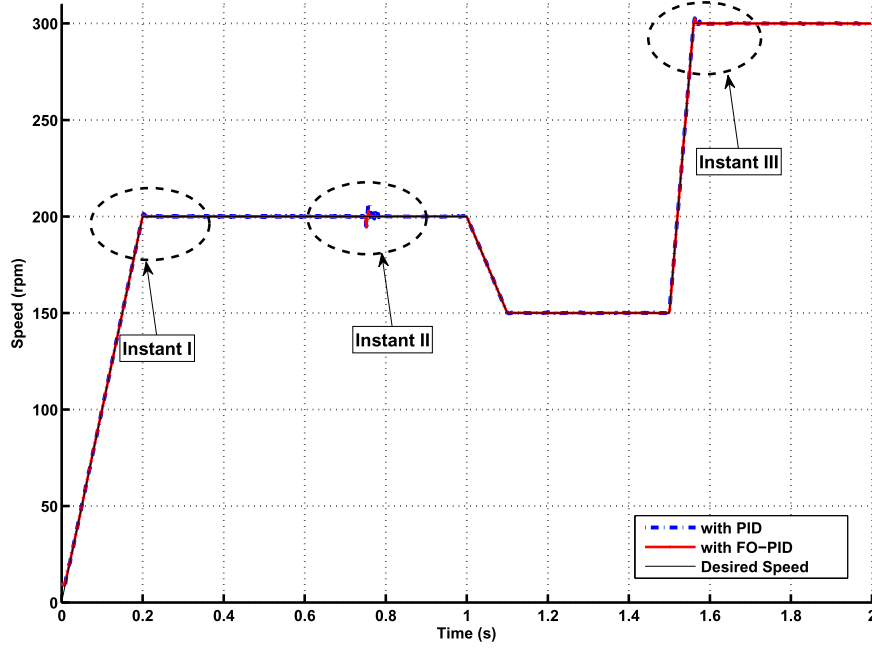


Figure 3: Speed response of PID and FO-PID controlled buck converter fed PMDC motor

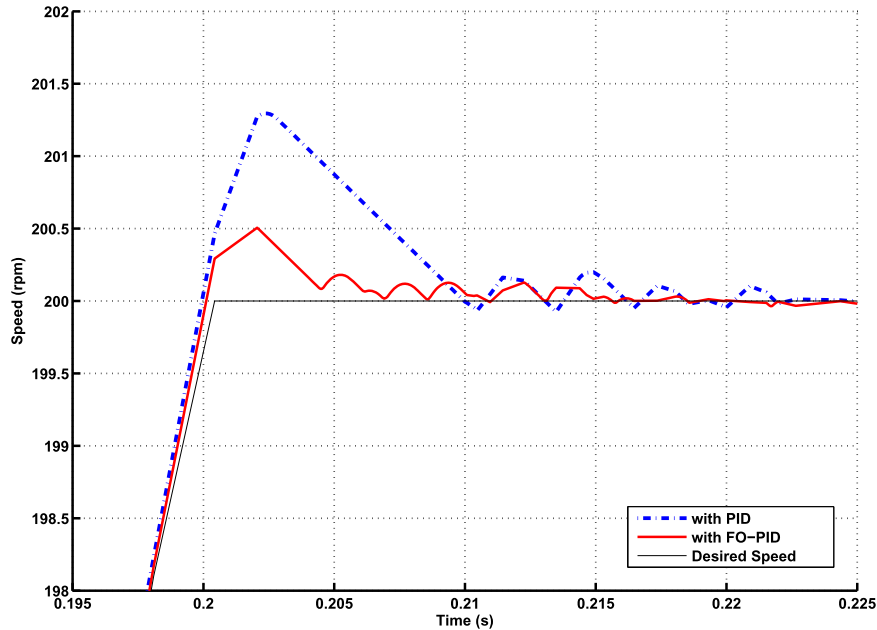


Figure 4: Zoom view of instant I

is obtained as (12),

$$u_d(z) = u(s'), \quad \text{where } s' = \frac{2}{T_s} \frac{z-1}{z+1} \quad (13)$$

3.2 Tuning Methodology

Ant colony optimization (ACO) is a technique for approximate optimization [39–42]. In the artificial intelligence (AI) perspective, ACO algorithms are one of the

most successful strands of swarm intelligence [40,41]. The goal of optimization is the design of intelligent multi-agent systems by taking inspiration from the collective behavior of social insects such as ants, bees, and other animal societies such as flocks of birds or fish schools. Examples of AI algorithms other than ACO are those for clustering and data mining inspired by ant's cemetery building behavior, those for dynamic task allocation inspired by the behavior of wasp colonies [39–41,43].

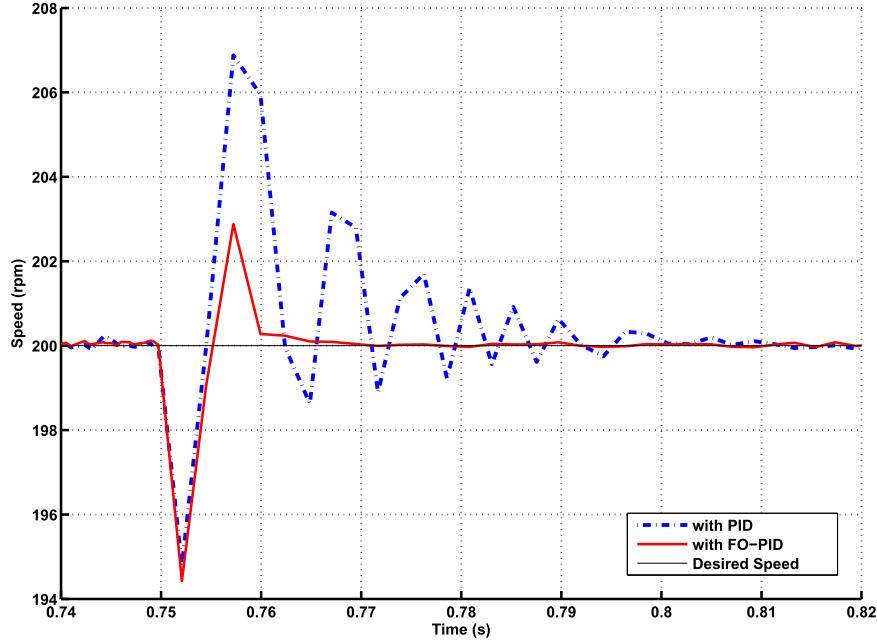


Figure 5: Zoom view of instant II

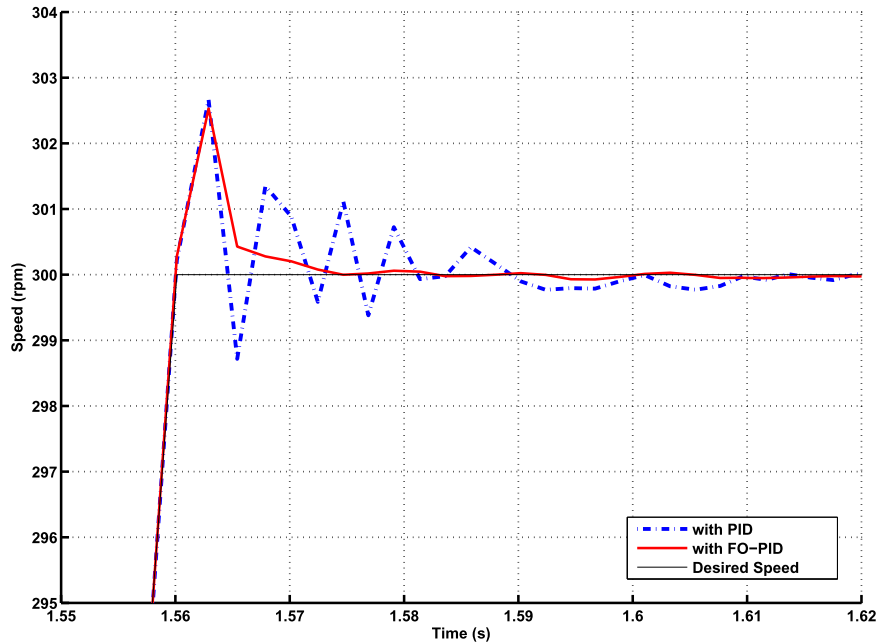


Figure 6: Zoom view of instant III

ACO is an algorithm for finding optimal paths that are based on the behavior of ants searching for food [42]. In the natural world, ants wander randomly and upon finding a source of food they return to their colony while laying down pheromone trails that show the path has food. If other ants come across such a path, they are likely to follow the path with a certain probability. If they do so, they populate the path with their own markers as they bring the food back. Over the time, the pheromone trail starts to evaporate, thus, reducing its attractive strength. The more time it takes for an ant to travel down the path and back again, the more time the pheromone has to evaporate. Because the ants drop pheromone every time they bring food, shorter paths are more likely to be stronger than longer ones, hence optimizing the solution.

In the meantime, some ants are still randomly scouting for closer food sources. Once the food source is depleted, the route is no longer populated with pheromone and slowly decays. Here, ACO algorithm is used for optimal tuning of FO-PID controller parameters by minimizing the required cost function (integral time absolute error). The ACO algorithm works as follows: In general, the k th ant moves from state i to state j with probability as given in (14),

$$p_{ij}^k(t) = \frac{\tau_{ij}(t)^\gamma (\frac{1}{d_{ij}})^\nu}{\sum_{j \in \text{nodes}} \tau_{ij}(t)^\gamma (\frac{1}{d_{ij}})^\nu} \quad (14)$$

where p is the probability, τ is the quantity of pheromone between the state i to state j , d heuristic distance between

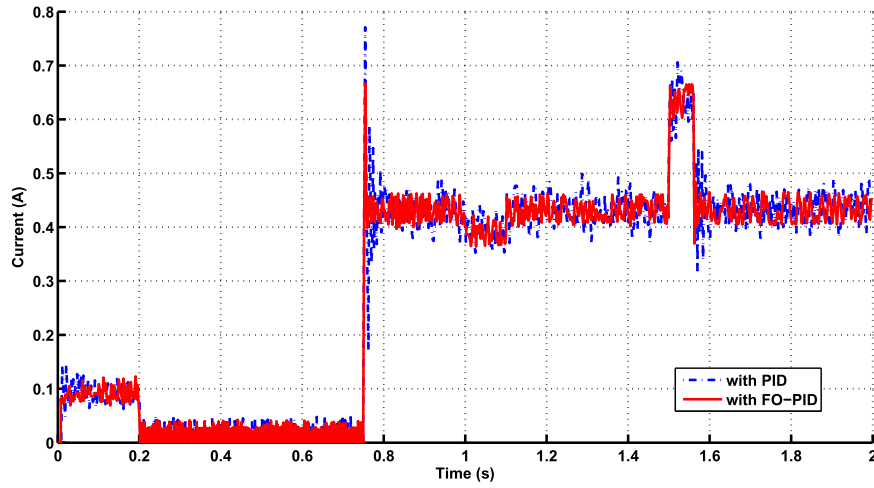


Figure 7: Armature current plot of PID and FO-PID controlled buck converter fed PMDC motor

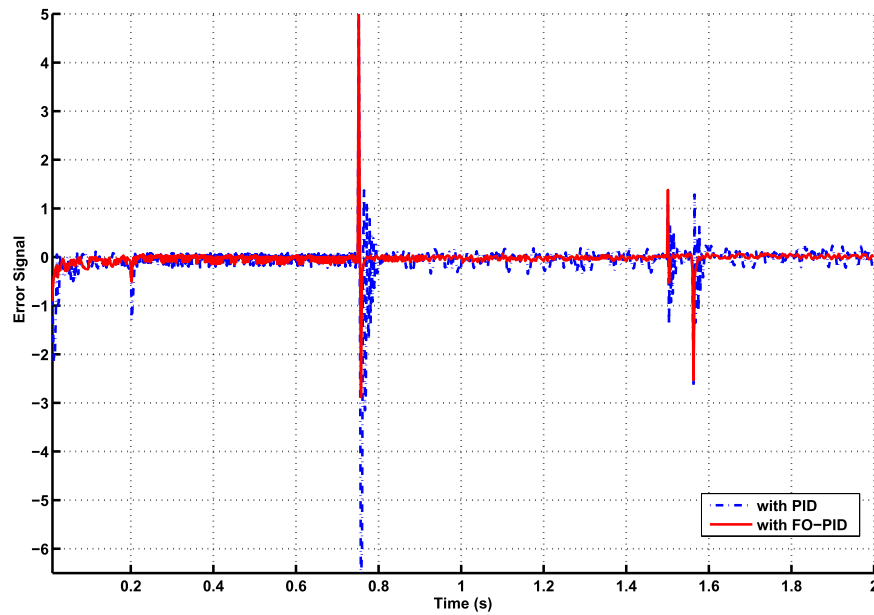


Figure 8: Error signal plot of buck converter fed PMDC motor

the state i to state j . The parameters γ and ν tune the relative importance in probability of the amount of pheromone versus the heuristic distance. When all the ants have completed a solution, the trails are updated by (15),

$$\tau_{ij}(t+1) = (1 - \rho)\tau_{ij}(t) + \sum_{k \in \text{colony that used edge}(i,j)} \frac{Q}{L_k} \quad (15)$$

where Q is a constant parameter, L_k is the k th ant solution, ρ is a coefficient representing pheromone evaporation, k is the number of ants. The number of ants is taken as 100,

Table 2: Error signal indices

Sr. no.	Controllers	ITAE
1	PID	0.3047
2	FO-PID	0.0535

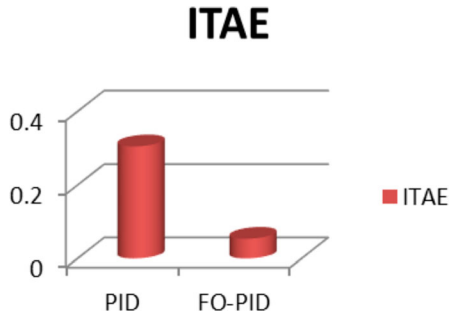


Figure 9: Error signal analysis

number of iterations is set as 50, number of parameters is taken as 5, the lower and higher translation frequencies are taken as $\omega_L = 0.1$ rad/s and $\omega_H = 1000$ rad/s. γ , ν , and ρ values are 0.8, 0.2, and 0.7 respectively [44]. Integral time absolute error (ITAE) is taken as a cost function [45–48]. The ITAE cost function “ J_{ITAE} ” is given in (16),

$$J_{ITAE} = \min \left(\int_0^\infty t|e(t)| dt \right) \quad (16)$$

where $e(t)$ is the difference between the actual value and specified value. The values of the controller parameters are obtained using the ACO algorithm and given in Table 1. These parameters are used in the simulation.

3.3 FO-PID Controller Design Steps

- (1) Band of frequency (ω_L, ω_H) is noted from open-loop Bode plot of the plant.
- (2) K_p, K_i, K_d, α , and β values are obtained using the ACO algorithm.
- (3) The controller transfer function is computed using (9).

$$c(s) = 12.06 + 16.37s^{-0.6} + 0.65s^{0.5} \quad (17)$$

- (4) Equation (17) is approximated using Oustaloup *et al.* method as in (10) and then discretized using

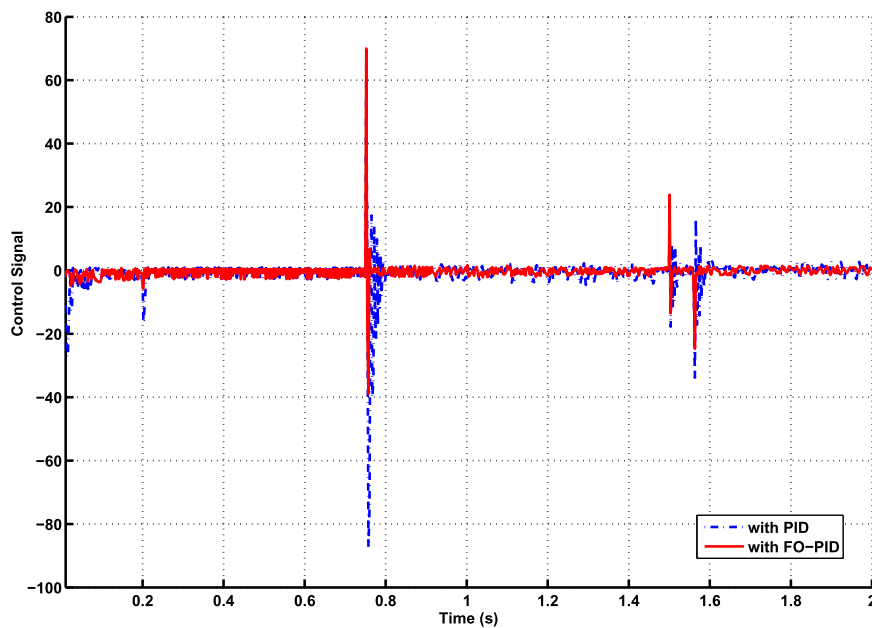


Figure 10: Control signal plot of buck converter fed PMDC motor

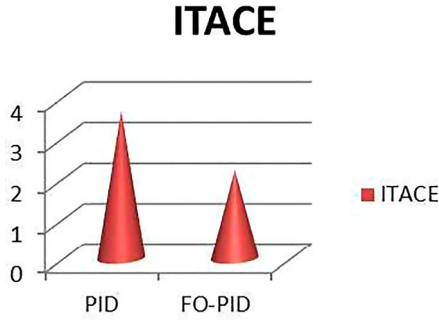
Table 3: Control signal indices

Sr. no.	Controllers	ITAE
1	PID	3.655
2	FO-PID	2.204

“Tustin” method using (12).

$$\frac{u_d(z)}{e(z)} = \frac{155.5z^{17} - 1143z^{16} + 3169z^{15} - 3083z^{14} - 3220z^{13} + 1.077e04z^{12} - 7497z^{11} - 5522z^{10} + 1.16e04z^9 - 4338z^8 - 3916z^7 + 4078z^6 - 614.8z^5 - 772.9z^4 + 366z^3 + 3.046z^2 - 33.52z + 5.861}{z^{17} - 8.439z^{16} + 28.89z^{15} - 46.25z^{14} + 15.08z^{13} + 69.65z^{12} - 116.8z^{11} + 48.81z^{10} + 62.01z^9 - 88.66z^8 + 30.3z^7 + 19.85z^6 - 21.91z^5 + 6.104z^4 + 1.305z^3 - 1.252z^2 + 0.2941z - 0.02317} \quad (18)$$

- (5) The approximated controller transfer function as in (18) is used in control scheme as speed controller and the performance of buck converter fed PMDC motor is observed.

**Figure 11: Control signal analysis**

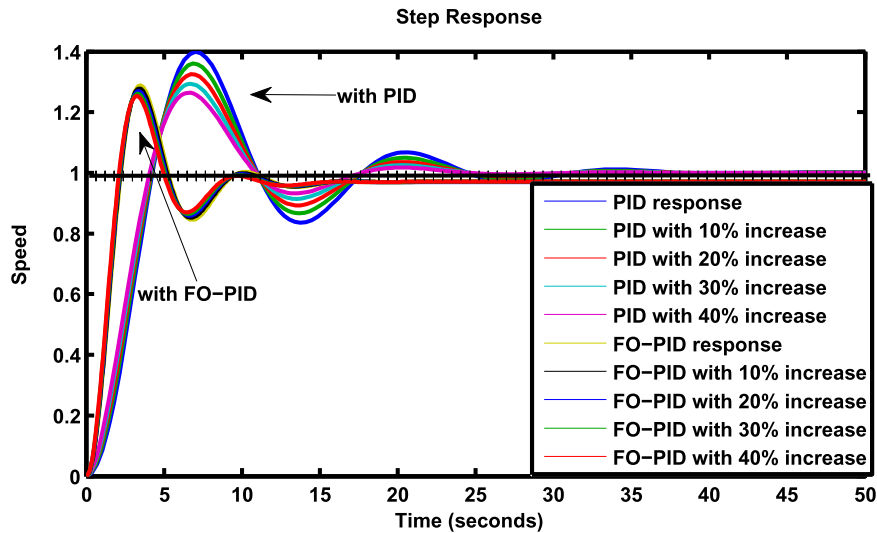
4. RESULTS

Speed response of buck converter fed PMDC motor at various set speed conditions is observed. The load is applied at the instant of 0.75 s from the initial condition. The speed response of PID and FO-PID controlled buck converter fed PMDC motor is presented in Figure 3.

The improved speed is observed in the case of the FO-PID controller and the current consumed by the FO-PID controller is less as compared to the PID controller. The magnified view of Figure 3 at instants I, II, and III is shown in Figures 4, 5, and 6 respectively. The armature current of PID and FO-PID controlled buck converter fed PMDC motor is presented in and Figure 7.

Figure 8 illustrates the error signal of PID and FO-PID controlled buck converter fed PMDC motor. The performance indices for error signal are presented in Table 2 and graphically presented in Figure 9. It is observed that the integral time absolute error (ITAE) is less in the case of the FO-PID controller. It infers that error is optimum in the case of FO-PID controller.

The control signal (the signal which comes out from the controller) plot for PID and FO-PID controlled buck converter fed PMDC motor is presented in Figure 10. Its analysis is shown in Table 3 and graphically presented in Figure 11. It is observed that the control efforts taken by the FO-PID controller are very less compared to its integer counterpart. Control efforts is quantized using performance indices integral time absolute control effort (ITACE).

**Figure 12: Unit step response of PID and FO-PID up to 40% increase in gain value**

For the performance analysis of PID and FO-PID controllers, DC forward path gain is increased up to 40% and corresponding responses are observed in Figure 12. It shows the robustness in parameter variation with the proposed FO-PID controller.

5. CONCLUSION

This paper presents the design and tuning of FO-PID-based speed controller for buck converter fed PMDC motor. A digital FO-PID controller is successfully implemented. Ant colony optimization is used for obtaining the FO-PID controller parameters. It is observed that the proposed ACO optimized digital FO-PID controller reduces the maximum overshoot, settling time, and ITAE as compared to the integer-order PID controller. The results infer that improvement in speed control is achieved with lesser control efforts using a proposed controller. The current required by the FO-PID controller is less as compared to the PID controller which implies an energy-efficient nature of the proposed controller. This control scheme can be applied to robotics and servo mechanisms.

DISCLOSURE STATEMENT

No potential conflict of interest was reported by the author(s).

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REFERENCES

1. J. P. Jiang, S. Chen, and P. K. Sinha, "Optimal feedback control of direct current motors," *IEEE Trans. Ind. Electron.*, Vol. 31, pp. 269–74, 1990.
2. V. Subrahmanyam, *Electric Drives: Concepts and Applications*. Noida: Tata McGraw-Hill Education, 2011.
3. R. S. Ortigoza, V. M. H. Guzman, M. A. Cruz, and D. M. Carrillo, "DC/DC buck power converter as a smooth starter for a DC motor based on a hierarchical control," *IEEE Trans. Power Electron.*, Vol. 30, no. 2, pp. 1076–84, 2015.
4. H. El Fadil and F. Giri, "Accounting of DC-DC power converter dynamics in DC motor velocity adaptive control," in *Proceedings of the IEEE International Conference on Control Applications*, Munich, Germany, Oct. 4–6, 2006, pp. 3157–62.
5. M. A. Ahmad, R. M. T. Raja Ismail, and M. S. Ramli, "Control strategy of buck converter driven DC motor: A comparative assessment," *Aust. J. Basic Appl. Sci.*, Vol. 4, no. 10, pp. 4893–903, 2010.
6. M. Z. M. Tumari, M. S. Saealal, M. R. Ghazali, and Y. A. Wahab, "H-infinity with pole placement constraint in LMI region for a buck-converter driven DC motor," in *Proceedings of the IEEE International Conference on Power and Energy*, Kota Kinabalu Sabah, Malaysia, Dec. 2–5, 2012, pp. 530–5.
7. P. Dhinakaran and D. Manamalli, "Novel strategies in the model-based optimization and control of permanent magnet DC motors," *Comput. Electr. Eng.*, Vol. 44, pp. 34–41, 2015.
8. H. Sira-Ramirez and M. A. Oliver-Salazar, "On the robust control of buck-converter DC motor combinations," *IEEE Trans. Power Electron.*, Vol. 28, no. 8, pp. 3912–22, 2013.
9. K. Astrom, and T. Hagglund, *PID Controllers: Theory, Design, and Tuning*. North Carolina: Instrument Society of America, 1995.
10. I. Podlubny, *Fractional Differential Equations*. New York: Academic Press, 1999.
11. S. Das, *Functional Fractional Calculus*, 2nd ed. Springer Science and Business Media, 2011.
12. I. Petras, *Fractional Order Non Linear Systems: Modeling, Analysis and Simulation*. Beijing and Berlin: Higher Education Press and Springer-Verlag, 2011.
13. S. Sutha, P. Lakshmi, and S. Sankaranarayanan, "Fractional-order sliding mode controller design for a modified quadruple tank process via multi-level switching," *Comput. Electr. Eng.*, Vol. 45, pp. 10–21, 2015.
14. A. Rebai, K. Guesmi, D. Gozim, and B. Hemici, "Design of an optimized fractional order fuzzy PID controller for a piezoelectric actuator," *J. Control Eng. Appl. Informat.*, Vol. 17, no. 3, pp. 41–9, 2015.
15. J. Devate, S. Y. Kulkarni, and K. R. Pai, "Wideband differentiator and integrator with ideal phase response," *Can. J. Electr. Comput. Eng.*, Vol. 38, no. 4, pp. 294–9, 2015.
16. I. Petras, "Fractional-order feedback control of a DC motor," *J. Electr. Eng.*, Vol. 60, no. 3, pp. 117–28, 2009.
17. V. Kumar, K. P. S. Rana, and P. Mishra, "Robust speed control of hybrid electric vehicle using fractional order fuzzy PD and PI controllers in cascade control loop," *J. Franklin Institute*, Vol. 353, no. 8, pp. 1713–41, 2016.
18. A. S. Chopade, S. W. Khubalkar, A. S. Junghare, M. V. Aware, and S. Das, "Design and implementation of digital fractional order PID controller using optimal pole-zero approximation method for magnetic levitation system," *IEEE/CAA J. Autom. Sin.*, Vol. 5, no. 5, pp. 977–89, 2018.

19. S. W. Khubalkar, A. S. Chopade, A. S. Junghare, and M. V. Aware, "Design and tuning of fractional order PID controller for speed control of permanent magnet brushless DC motor," in *IEEE First International Conference on Control, Measurement and Instrumentation (CMI)*, 2016, pp. 326–330.
20. S. Khubalkar, A. Junghare, M. Aware, and S. Das, "Modeling and control of a permanent-magnet brushless DC motor drive using a fractional order proportional-integral-derivative controller," *Turk. J. Electr. Eng. Comput. Sci.*, Vol. 25, no. 5, pp. 4223–41, 2017.
21. C. A. Monje, B. M. Vinagre, V. Feliu, and Y. Q. Chen, "Tuning and auto-tuning of fractional order controllers for industry applications," *Control Eng. Pract.*, Vol. 16, pp. 798–812, 2008.
22. F. Padula and A. Visioli, "Optimal tuning rules for proportional-integral derivative and fractional-order proportional-integral derivative controllers for integral and unstable processes," *IET Control Theory Appl.*, Vol. 6, no. 6, pp. 776–86, 2012.
23. R. Duma, M. Trusca, and P. Dobra, "Tuning and implementation of PID controllers using rapid control prototyping," *J. Control Eng. Appl. Informat.*, Vol. 13, no. 4, pp. 64–73, 2011.
24. W. Mitkowski and K. Oprzedkiewicz, "Tuning of the half-order robust PID controller dedicated to oriented PV system," in *Advances in Modeling and Control of Non-integer Order Systems*, Lecture Notes in Electrical Engineering, Vol. 320, 2015, pp. 145–57.
25. R. C. Dorf and R. H. Bishop, *Modern Control Systems*, 12th ed. Pearson, 2011.
26. N. Mohan, T. M. Undeland, and W. P. Robbins, *Power Electron*, 3rd ed. Wiley, 1988.
27. S. G. Kadwane, S. P. Vepa, B. M. Karan, and T. Ghose, "Converter based DC motor speed control using TMS320LF2407A DSK," in *Proceedings of the IEEE International Conference on Industrial Electronics and Application*, Singapore, May 24–26, 2006, pp. 1–5.
28. M. O. Efe, "Fractional order systems in industrial automation – A survey," *IEEE Trans. Ind. Informat.*, Vol. 7, no. 4, pp. 582–91, 2011.
29. I. Pan and S. Das, "Gain and order scheduling for fractional order controllers," in *Intelligent Fractional Order Systems and Control*, Chapter 6, Vol. 438. Berlin: Springer-Verlag, 2013, pp. 147–57.
30. D. Chen, Y. Q. Chen, and D. Xue, "Digital fractional order Savitzky-Golay differentiator," *IEEE Trans. Circuits Syst. II, Exp. Brief*, Vol. 58, no. 11, pp. 758–62, 2011.
31. J. A. T. Machado, "Discrete-time fractional-order controllers," *Fract. Calc. Appl.*, Vol. 4, no. 1, pp. 47–66, 2001.
32. Y. Q. Chen and K. L. Moore, "Discretization schemes for fractional order differentiators and integrators," *IEEE Trans. Circuits Syst. I Fundam. Theory Appl.*, Vol. 49, no. 3, pp. 363–7, 2002.
33. I. Podlubny, "Fractional-order systems and $PI^\lambda D^\mu$ -controllers," *IEEE Trans. Autom. Control*, Vol. 44, no. 1, pp. 208–14, 1999.
34. D. Valerio and J. Sa Da Costa, "Introduction to single-input, single output fractional control," *IET Control Theory Appl.*, Vol. 5, no. 8, pp. 1033–57, 2011.
35. P. Lanusse, J. Sabatier, and A. Oustaloup, "Fractional order PID and first generation crone control system design," in *Fractional Order Differentiation and Robust Control Design*, Intelligent Systems, Control and Automation: Science and Engineering, Chapter 2, Vol. 77, 2015, pp. 63–105.
36. B. M. Vinagre, I. Podlubny, A. Hernandez, and A. Feliu, "Some approximations of fractional order operators used in control theory and applications," *Fract. Calc. Appl. Anal.*, Vol. 3, no. 3, pp. 231–48, 2000.
37. A. Charef, H. H. Sun, Y. Y. Tsao, and B. Onaral, "Fractal system as represented by singularity function," *IEEE Trans. Autom. Control*, Vol. 37, no. 9, pp. 1465–70, 1992.
38. A. Oustaloup, F. Levron, B. Mathiew, and F. Nanot, "Frequency band complex noninteger differentiator: Characterization and synthesis," *IEEE Trans. Circuits Syst. I Fundam. Theory Appl.*, Vol. 47, no. 1, pp. 25–39, 2000.
39. C. Blum, "Ant colony optimization: Introduction and recent trends," *Phys. Life Rev.*, Vol. 2, pp. 353–73, 2005.
40. M. Dorigo, M. Birattari, and T. Stutzle, "Ant colony optimization: Artificial ants as a computational intelligence technique," *IEEE Comput. Intell. Mag.*, Vol. 1, pp. 28–39, 2006.
41. M. Dorigo, V. Maniezzo, and A. Colorni, "Ant system: Optimization by a colony of cooperating agents," *IEEE Trans. Syst. Man Cybern. A Syst. Humans*, Vol. 26, no. 1, pp. 29–41, 1996.
42. E. Bonabeau, M. Dorigo, and G. Theraulaz, *Swarm intelligence: From natural to artificial systems*. New York: Oxford University Press, 1999.
43. J. Kennedy and R. C. Eberhart, "Particle swarm optimization," in *IEEE International Conference on Neural Networks*, Vol. 4. Piscataway, NJ: IEEE Press, 1995.
44. I. Chaouch, O. B. Driss, and K. Ghedira, "A modified ant colony optimization algorithm for the distributed job shop scheduling problem," *Procedia Comput. Sci.*, Vol. 112, pp. 296–305, 2017.
45. E. Poulin and A. Pomerleau, "PID tuning for integrating and unstable processes," *IEE Proc. Control Theory Appl.*, Vol. 143, no. 5, pp. 429–35, 1996.

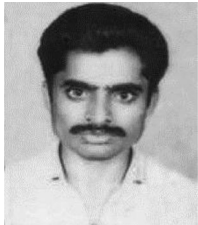
46. A. K. Maurya, M. R. Bongulwar, and B. M. Patre, "Tuning of fractional order PID controller for higher order process based on ITAE minimization," in *Annual IEEE India Conference (INDICON)*, 2015, pp. 1–5.
47. F. G. Martins, "Tuning PID controllers using the ITAE criterion," *Int. J. Eng. Educ.*, Vol. 21, no. 5, pp. 1–7, 2005.
48. S. W. Sung, J. Lee, and I. B. Lee, *Process Identification and PID Control*, Vol. 6. Singapore: Wiley, 2009.

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